

## Product Requirements Document (PRD) — AxonAI

- Version: 1.0
- Date: 2025-08-26
- Product: AxonAI — AI-driven personal productivity and scheduling assistant
- Owners: Product + Eng
- Stakeholders: Eng, Design, Security, GTM

### 1) Overview

- **Problem:** Knowledge workers struggle to prioritize, plan, and protect focused time across fragmented task systems (Jira, monday.com, calendars). Manual planning is time-consuming and brittle.
- **Solution:** AxonAI centralizes tasks and time signals, then uses AI to generate plans, schedules, and insights (e.g., burnout risk, efficiency trends), with native integrations to Jira and monday.com.
- **Primary KPIs**
- **planning-adoption:** weekly active users generating an AI plan  $\geq 2x/week$
- **schedule-commitment:** % of scheduled tasks completed on-time
- **burnout-alerts-ack:** % of burnout alerts acknowledged within 24h
- **integration-coverage:** % users with  $\geq 1$  external integration connected
- **Release Target:** Public Beta

### 2) Goals and Non-Goals

- **Goals**
- **AI planning** from task lists and historical time usage.
- **Calendar-aware schedules** that respect meetings/availability.
- **Integrations** with Jira and monday.com (OAuth, status, task sync).
- **Insights:** time usage, efficiency score, burnout prediction.
- **Meeting prep** suggestions and pre-reads.
- **Task breakdown** and dynamic reallocation.
- **Notifications** via email for plans, alerts, and changes.
- **Non-Goals**

- Full-blown project portfolio management (enterprise Gantt).
- Direct calendar provider write-back in v1 (read-only assumed).
- Fine-grained org permissions/roles (team admin console).
- Financial/time-sheet compliance.

### 3) Personas

- **Individual Contributor (IC):** Engineers/designers/analysts optimizing daily work.
- **Team Lead/PM:** Needs visibility into workload balance and risk signals.
- **Startup Founder:** Maximizes output; rapidly plans around meetings and context switches.

### 4) Key Use Cases

- **Plan my day:** From imported tasks, generate a focused plan with time blocks.
- **Break down a big task:** Create actionable subtasks with estimates and priorities.
- **Reallocate mid-day:** Adjust remaining schedule after unplanned work/meetings.
- **Meeting prep:** Generate pre-read summary and talking points.
- **Connect Jira/monday:** Pull tasks, statuses; push updates (as supported).
- **Monitor burnout:** Detect elevated risk and recommend mitigation strategies.

### 5) Scope and Features

- **Dashboard (src/app/(app)/dashboard/page.tsx)**
- Shows today's plan, quick actions, and key insights (efficiency, burnout).
- Priority: P0
- **Tasks (src/app/(app)/tasks/page.tsx, src/components/tasks/\*)**
- View/import tasks; mark complete; AI "plan from tasks".
- Priority: P0
- **Schedule (src/app/(app)/schedule/create/page.tsx, src/components/schedule/create-schedule-form.tsx)**
- AI creates calendar-aware schedule using meetings/availability.

- Priority: P0
- **Reallocate**  
(src/app/(app)/reallocate/page.tsx, src/components/reallocate/reallocate-form.tsx)
- Re-plan remaining time based on actuals/interruptions.
- Priority: P1
- **Breakdown**  
(src/app/(app)/breakdown/page.tsx, src/components/breakdown/breakdown-form.tsx)
- Intelligent task decomposition with estimates.
- Priority: P1
- **Analytics** (src/app/(app)/analytics/page.tsx, src/components/analytics/\*)
- Time usage, progress, efficiency score, burnout prediction.
- Priority: P1
- **Meeting Prep** (src/app/(app)/meeting-prep/page.tsx, src/components/meeting/meeting-prep-form.tsx)
- Agenda, pre-read summarization, action prompts.
- Priority: P2
- **Talk to Founder** (src/app/(app)/talk-to-founder/page.tsx)
- Feedback channel.
- Priority: P3
- **Settings** (src/app/(app)/settings/page.tsx, src/components/settings/\*)
- Theme, integrations (Jira, monday), notifications.
- Priority: P0
- **Auth** (src/app/(auth)/\*, src/components/auth/auth-provider.tsx)
- Email/password or provider (actual method TBD by implementation).
- Priority: P0
- **AI Flows** (src/ai/flows/\*, src/ai/genkit.ts)
- analyze-time-usage, calculate-efficiency-score, create-schedule,

dynamic-task-reallocation, intelligent-task-breakdown, predict-burnout, speech-meeting-aware, chat.

- Priority: P0-P1
- **APIs (Next.js App Router, src/app/api/\*)**
- Chat: api/chat/route.ts
- AI planning: api/ai/plan-for-day, api/ai/plan-from-tasks
- Jira: api/jira/{auth,callback,status,tasks}
- Monday: api/monday/{auth,callback,status,tasks,webhook,webhooks/{register,delete}}
- Notifications: api/notify/email
- Priority: P0

## 6) Functional Requirements

- **FR1: AI Plan Generation**
  - Input: tasks (title, duration estimate, priority, tags), user availability, meetings.
  - Output: ordered plan with time blocks; rationale; confidence.
  - Constraints: must respect time windows and maximum daily load.
- **FR2: Calendar-aware Scheduling**
  - Reads meetings, avoids overlaps, preserves buffers.
  - Supports reallocation with minimal disruption.
- **FR3: Integrations**
  - Jira and monday.com OAuth flows; status endpoints must reflect connectivity.
  - Task fetch: minimally title, status, assignee, estimate, due date, url.
  - Idempotent webhook registration/deletion for monday.com.
- **FR4: Analytics and Insights**
  - Time usage breakdown by category; weekly trends.
  - Efficiency score definition (v1 heuristic): ratio of planned vs completed on time, adjusted for focus depth.
  - Burnout signal from workload trend, after-hours load, meeting load.
- **FR5: Meeting Prep**

- Summarize context; suggest agenda; pre-populate notes.
- **FR6: Notifications**
- Email notifications for new plans, schedule changes, burnout alerts.
- **FR7: Security and Auth**
- Session-based auth; protect all app routes under (app) layout.
- Tokens/credentials stored securely on server; do not expose in client.

## 7) Non-Functional Requirements

- **Performance**
- P50 AI endpoints < 3s; P95 < 8s.
- Page TTI < 2.5s on median hardware.
- **Availability**
- 99.5% monthly for core APIs.
- **Scalability**
- Support 10K MAU without architecture change.
- **Privacy/Security**
- Store minimal PII; encrypt at rest/in transit; configurable data retention.
- OAuth secrets via server-side secrets store (e.g., env vars/secret manager).
- Least-privilege scopes for Jira/monday.

## 8) Information Architecture

- **Primary**  
**Navigation:** dashboard, tasks, calendar, analytics, reallocate, breakdown, meeting-prep, settings, help, talk-to-founder.
- **Global UI:** app-header, app-sidebar, mobile-tabbar, ai-action-button.
- **UI Library:** Shadcn UI/Radix components (src/components/ui/\*) and Tailwind.

## 9) Data Model (logical)

- **User**
- id, email, displayName, settings (theme, work hours), integration tokens.
- **Task**

- id, source (manual, jira, monday), title, status, estimate, dueDate, priority, url, tags.
- **Plan**
- id, userId, date, items: [{taskId, start, end, confidence, notes}]
- **Schedule**
- generated from plan; immutable history of versions.
- **AnalyticsMetrics**
- timeUsage, efficiencyScore, burnoutRisk, computedAt.

Storage backing implied via src/lib/\* (Firebase present).

## 10) AI/ML Requirements

- **Model Orchestration:** via src/ai/genkit.ts.
- **Determinism:** low-temp defaults, include rationale in outputs for transparency.
- **Guardrails:** enforce business constraints (work hours, max daily load).
- **Evaluation:** weekly review against KPIs and user feedback.

## 11) API Contracts (high level)

- POST /api/ai/plan-for-day
- Body: { tasks, availability, meetings }
- Result: { plan: Plan, rationale, confidence }
- POST /api/ai/plan-from-tasks
- Body: { tasks }
- Result: { planDraft }
- POST /api/chat
- Body: { messages[] }
- Result: { message }
- GET /api/{jira|monday}/status
- Result: { connected: boolean, details }
- GET /api/{jira|monday}/tasks
- Query: paging

- Result: { tasks[] }
- POST /api/notify/email
- Body: { to, subject, html|text }
- POST /api/monday/webhooks/register|delete
- Body: { boardId, callbackUrl } | { webhookId }

## 12) UX Requirements

- **Accessibility:** WCAG AA; keyboard nav; ARIA attributes.
- **Responsive:** mobile-first with mobile-tabbar.
- **Consistency:** use ui/\* primitives; theme toggle in settings.

## 13) Telemetry & Analytics

- Event taxonomy:
- plan\_generated, schedule\_applied, task\_completed\_on\_time, reallocated\_midday, burnout\_alert\_shown, integration\_connected, meeting\_prep\_generated.
- Error tracking on API routes with correlation ids.

## 14) Rollout Plan

- Alpha (internal): core AI planning, tasks, schedule, Jira auth.
- Beta (early users): monday.com integration, analytics, burnout, notifications.
- GA: meeting prep, reallocation, webhook automation, performance polish.

## 15) Risks & Mitigations

- Model hallucinations → constrain prompts; validate outputs; add guardrails.
- Integration rate limits → caching, backoff, minimal scopes.
- Privacy concerns → clear consent, data minimization, configurable retention.
- Scheduling conflicts → strict overlap checks; user override UX.

## 16) Success Metrics

- D1 retention of planners > 35%
- Weekly plan generation per active user ≥ 3
- On-time completion rate uplift ≥ 20% over baseline
- ≥ 50% users connect at least one integration

## 17) Open Questions

- Calendar write-back in v1 or v2?
- Which auth providers beyond email/password?
- How are time sources ingested for “actuals” if calendar-only is insufficient?

If you want, I can save this as docs/prd.md and open a PR.

10%

3 To-dos

Add Context